



GAS (Grover's Algorithm Search)

Source: Grover's algorithm in a four-qubit silicon processor above the fault-tolerant threshold

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Grover's Algorithm: Simulation vs Reality

- What is Grover's algorithm?
- Why it matters? (Finance)
- My Objective



Level 1

How Grover's Algorithm Works

Uses **amplitude amplification** to increase probability of correct state

- Repeated application of:
 - Oracle (marks correct solution)
 - Diffusion operator (amplifies probability)
- Optimal number of iterations:

$$k \approx \frac{\pi}{4} \sqrt{N}$$

- Measurement returns target state with high probability



Method: Simulation Approach

- Implemented using Qiskit and AerSimulator
- Constructed Boolean oracle for a **random target state**
- Ran simulations from **2 to 20 qubits**
- Measured:
 - Runtime (seconds)
 - Success probability
 - Circuit depth
 - Gate count
- Circuits were **transpiled** to analyze real gate operations

Transpiled Circuit Representation

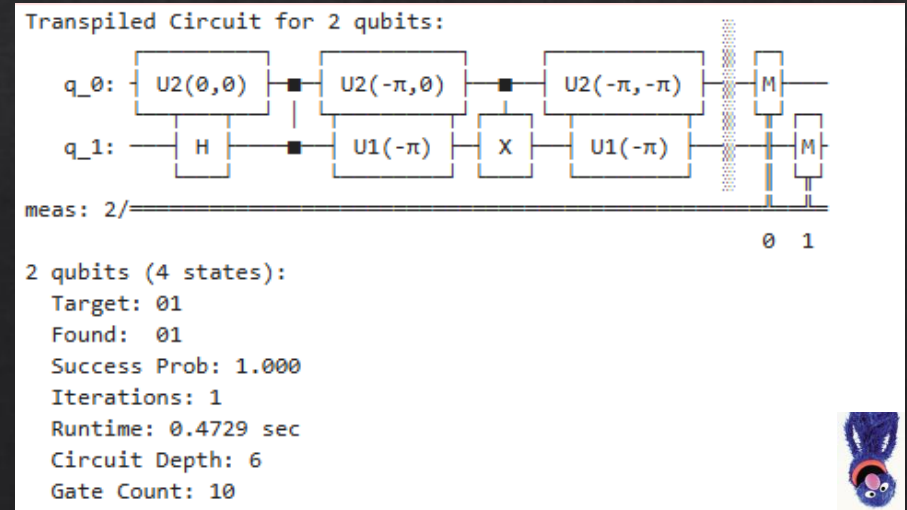
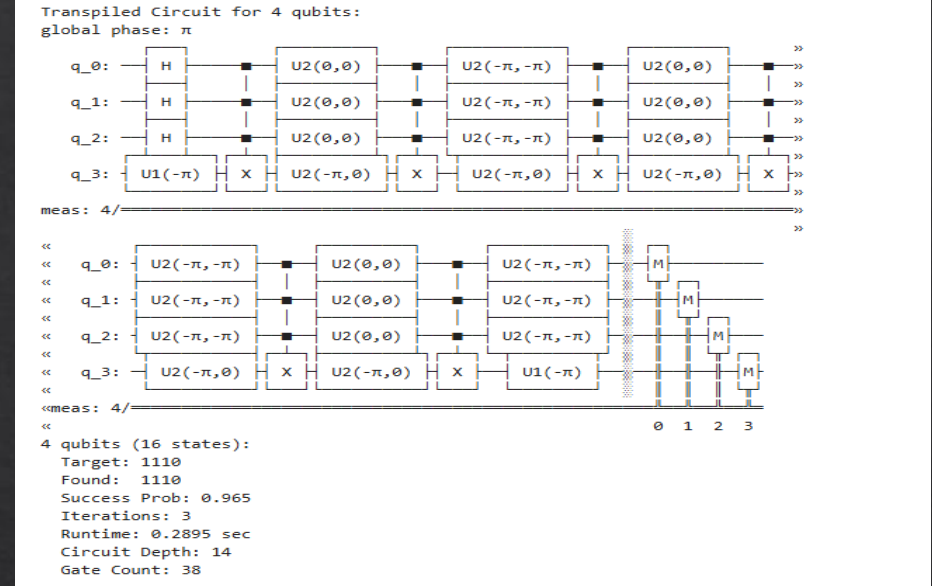
-Circuit decomposed into **basic quantum gates** after transpilation

Includes:

- Oracle (marks the solution)
- Diffusion operator (amplifies amplitude)
- Measurement step

Circuit depth and gate count increase with qubits

Greater depth $\rightarrow$ more operations $\rightarrow$ higher computational costs



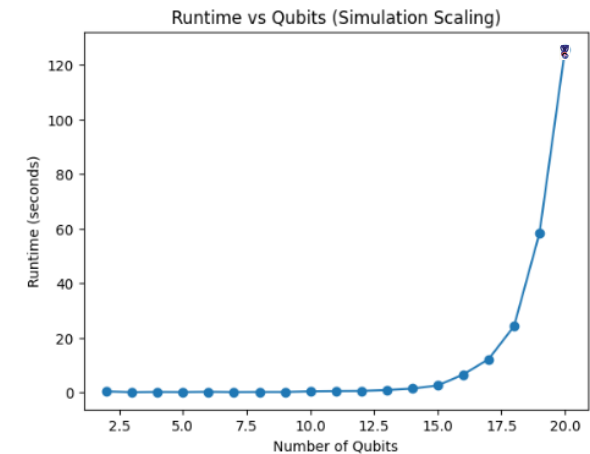
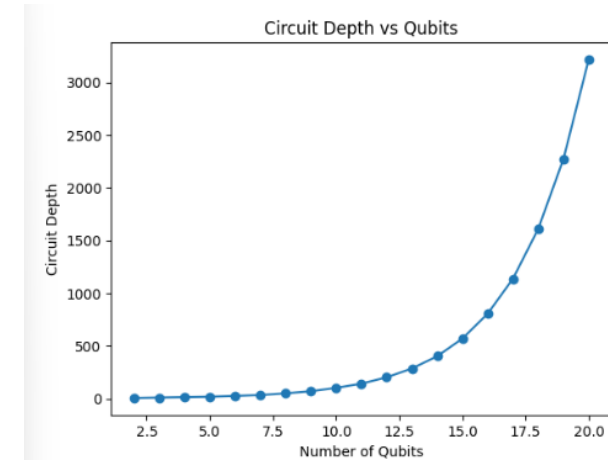
Results: Simulation Scaling

- Runtime increases rapidly as qubits increase
- -Circuit depth and gate count grow significantly
- Success probability remains ~ 1.0 for most cases

■ Example:

- 10 qubits $\rightarrow$ about 0.5 sec
- 15 qubits $\rightarrow$ about 2.6 sec
- 20 qubits $\rightarrow$ about 125 sec

- This shows an **exponential cost of classical simulation**



Conclusion and Key Takeaways

- Grover's algorithm successfully finds target states with high probability
- Classical simulation becomes **exponentially expensive** as qubits increase
 - Simulation must track 2^n quantum states
 - **Circuit complexity (depth and gates) drives runtime** 
- Real Quantum hardware is needed to achieve true speedup

DID you spot All the Grovers?